An update to this article is included at the end

Forest Ecology and Management 532 (2023) 120828

Contents lists available at ScienceDirect

Forest Ecology and Management

journal homepage: www.elsevier.com/locate/foreco

Faster evapotranspiration recovery compared to canopy development post clearcutting in a floodplain forest

T. Ghisi [a,b,*], M. Fischer [a,b], N. Kowalska [a], G. Jocher [a], M. Orság [a,b], M. Bláhová [a,b], H. Nieto [c], L. Homolová [a], Z. Žalud [a,b], M. Trnka [a,b]

[a] Global Change Research Institute of the Czech Academy of Sciences, Bělidla 986/4b, 603 00 Brno, Czech Republic
[b] Mendel University in Brno, Institute of Agrosystems and Bioclimatology, Zemědělská 1, 613 00 Brno, Czech Republic
[c] Institute of Agricultural Sciences-CSIC, Serrano 115 b, 28006 Madrid, Spain

ARTICLE INFO

Keywords:
Actual evapotranspiration
Clearcutting
Ecosystem recovery
Energy balance
Leaf area index
METRIC model
Remote sensing

ABSTRACT

Actual evapotranspiration (ET_a) is a key component of the water balance, and its accurate quantification of spatiotemporal variation is essential to improve the understanding of soil–plant-atmosphere-hydrosphere interactions. While in situ point measurements allow for the determination of the ET_a of a particular ecosystem or land cover type, remote sensing methods make it possible to divide and evaluate the rate of ET_a of different ecosystems in the landscape and would allow better management of the water resource system, especially if those are partly or permanently limited. The goal of this paper was to analyze the impact of clearcutting during standard forest management practices in floodplain forests and its effect on the water regime of the area. We selected the Lanžhot floodplain forest area (48.681°N 16.950°E, 155 m a.s.l.) that represents a biologically and hydrologically unique area of remnant floodplain forest within the upper Danube River basin. Since clearcutting represents a drastic intervention into the ecosystem including changes in canopy structure and local microclimate, we hypothesized that the impact of the clearcut on the overall water balance is long-term in the order of decades as a result consequence of logging of the mature forest and changes in local microclimatic conditions in disturbed areas. To analyze the impacts of clearcutting on ET_a, we applied the METRIC (Mapping EvapoTranspiration at high Resolution with Internalized Calibration) remote sensing model. METRIC was validated in the Central European climatic conditions using two ground eddy covariance measurements systems over grassland and floodplain forest. The evaluated model METRIC was then applied to quantify ET_a in three areas of interest in the floodplain forest, where in 2015–2021, mature vegetation was targeted for removal, modification of the soil surface, and subsequent planting of a new generation of English oak trees (Quercus robur L.). Immediately after the removal of vegetation cover, METRIC detected an annual decrease in ET_a by 35–40%, while leaf area index (LAI) decreased by 70–80%. This was followed by a quick recovery of the ET_a/ET_o (where ET_o represents FAO-56 reference evapotranspiration), reaching similar values as uncut surrounding vegetation during the third year, while LAI recovery was detected in the fifth year after the removal of vegetation cover. The rapid recovery of the ET_a/ET_o rejected the initial hypothesis, and studies showed that the newly formed stand replaced the ET_a/ET_o rate of the mature stand in the floodplain forest conditions of Central Europe in the fourth year, suggesting that the impacts of clearcutting on the hydrology may be temporally relatively limited.

1. Introduction

Removing vegetation cover, whether by natural or anthropogenic means, affects hydrological processes and the overall hydrological balance (Oulehle et al., 2021). These changes are reflected in the transpiration and evaporation capacity of the area, which follows the meteorological, pedological, and vegetation characteristics of an area of interest (AOI). For instance, Thompson et al. (2018) found that the groundwater level and runoff from the affected area increased after the removal of the aspen (*Populus tremula L.*) stand, but this increase was dependent on the gradual regeneration of the stand and its rate of actual evapotranspiration (ET_a). Stasik et al. (2020) recorded the impact of logging mature stands on raising the groundwater level in an ash-alder swamp forest in Poland. Watershed experiments in Japan

* Corresponding author.
E-mail address: ghisi.t@czechglobe.cz (T. Ghisi).

https://doi.org/10.1016/j.foreco.2023.120828
Received 28 November 2022; Received in revised form 23 January 2023; Accepted 25 January 2023
Available online 3 February 2023
0378-1127/© 2023 Elsevier B.V. All rights reserved.

demonstrated a reduction in the ET_a rate in the area by approximately 100 mm per year due to the logging of *Cryptomeria japonica*, with the reduced ET_a rate of the affected area being observed for the next twenty years (Kubota et al., 2021). Furthermore, Ide et al. (2013) reported that the effect of the clearcutting area on hydrological processes in a boreal forest in Finland persisted for 18 years after stand disturbance. It is assumed that the reduction in vegetation cover leads to a decrease in the transpiration rate of the area and an increase in runoff. On the other hand, there are observations from disturbed forests, which record only short-term changes in hydrological processes to a negligible extent (Beudert et al., 2018; Kopáček et al., 2020), or observable changes in the water balance are not recorded (Hruška et al., 2016). The rate of ET_a recovery in areas disturbed by logging, along with pedological and meteorological conditions, is influenced by the method of soil surface treatment and the method of gradual reforestation of the damaged areas. Studies by Sun et al. (2010), Yang et al. (2017) and Yang et al. (2020) indicated a positive linear relationship between the age of the tree and ET_a for pine stands. This linear relationship holds for approximately 20 years of tree age. Čermák and Prax (2009) confirmed in their study of floodplain forest in the Dolní Morava region that the transpiration of individual mature English oak trees significantly exceeds the transpiration of individual young trees.

ET_a represents the key component of the terrestrial water balance. Accurate ET_a analysis of the dynamically developing vegetation cover within anthropogenic or natural disturbances is key to determining the total water balance of the affected area. The current network density of in situ ET_a measurements is insufficient for a comprehensive ET_a picture of a heterogeneous landscape (Allen et al., 2007a). Remote sensing is one of the approaches to estimating the hydrological conditions on the surface within the hydrological balance in the landscape. The advantage of remote sensing methods is the ability to collect spatially distributed data on a global to local spatial scale. The physically based remotely sensed models based on the calculation of the surface energy balance of the AOI have been used for the spatial estimation of ET_a (Anderson et al., 1997; Bastiaanssen et al., 1998; Norman et al., 2003; Allen et al., 2007a; Senay et al., 2007).

The METRIC model (Allen et al., 2007a) is a remote sensing ET_a model capable of calculating regional ET_a based on the theory of surface energy balance. Studies in different countries of the world in other climatic conditions have confirmed a good agreement of ET_a outputs between the model and ground point ET_a measurement methods (Allen et al., 2007b; Hankerson et al., 2012; Mkhwanazi and Chávez 2012; Carrasco-Benavides et al., 2014; Zhang et al., 2015; Liebert et al., 2016). METRIC initially used shortwave and longwave (thermal) Landsat polar-orbiting satellite imagery. Trezza et al. (2013) applied METRIC using MODIS sensor. Recently, METRIC has started to be applied to high-resolution images taken by UAS (unmanned aircraft systems) sensors (Chandel et al., 2020) or onboard aircrafts. The model is internally calibrated using ET_o calculated from local weather data. METRIC uses the pioneer foundation of the SEBAL model (Bastiaanssen et al., 1998), which uses two endmember pixels ("dry and wet pixels") to solve the aerodynamic resistance and near-surface vertical temperature gradient between these two extreme points. This method greatly increases the calculation efficiency of the sensible heat flux.

In this study, the ET_a outputs from METRIC are evaluated using two ground-based eddy covariance (EC) measurements in the floodplain forest area in southern Moravia. After evaluating METRIC, the development and relationship of ET_a and LAI are analyzed in selected parts of the largest floodplain forest in Central Europe, where areas that experienced clearcutting were created due to targeted logging. The aim of this contribution is primarily to quantify the rate of ET_a, the ET_a/ET_o according to FAO, (Allen et al., 1998) and LAI depending on the targeted removal of mature vegetation cover and subsequently to detect their development in such damaged areas, including detecting the time of their possible recovery. Based on a heights study (Kubota et al., 2021), a significant decrease in the rate of ET_a and LAI is predicted in response to the removal of vegetation cover. According to others studies like Kubota et al. (2021) or Yang et al. (2017) who reported recovery of ET_a in about 20 years, we hypothesized that young vegetation cover cannot replace the level of ET_a and LAI of the extracted mature stand in the short term. We assumed that recovery of the ET_a, ET_a/ET_o, and LAI on the pre logging values is a long term process on the order of decades dependent on the development of the new vegetation cover.

This work will contribute to the clarification of changes in the individual components of the total water balance in response to a decrease in the LAI value during the targeted logging of mature vegetation cover and in the near post harvesting period in the conditions of Central European floodplain forests. Accurate knowledge of hydrological processes in various forest biomes is key for various ecosystem services of the forest and can thus significantly help management in these types of forests and can also help the management in these very rare species-diverse forest locations, which the floodplain forests in the lower Moravia region are. This study contributes to the water issue within forest management during the establishment and restoration of a forest within a floodplain ecosystem, whose ecosystem functions are key to ensuring ecological stability in the lower reaches of the Moravian and Pannonian biogeographic regions.

2. Materials and methods

2.1. Study area

The study area is located in the largest floodplain forest complex in the Czech Republic, located in the southernmost part of Morava near the confluence of the Morava and Thaya rivers near the city of Lanžhot (Fig. 1). The elevation of the entire area varies between 150 and 160 m. Fluvial soils are the dominant soil type. The three AOIs are forested areas where the mature forest of English oak and European ash *(Fraxinus excelsior L.)* (height of trees 25–35 m, age of trees 130–160 years) were harvested in 2016. This resulted in the creation of 14 clearcutting areas where reforestation is underway. (Lanžhot-Forest and Lanžhot-Grassland) are located in these areas. The ecosystem site Lanžhot-Forest contributes to ICOS (Integrated Carbon Observation System) network.

Forest clearcutting 1 (AOI 1) (Fig. 2) and 2 (AOI 2) (Fig. 3) were harvested at the end of 2016. Forest clearcutting 3 (AOI 3) (Fig. 4) occurred in 2016. Individual clearcutting areas are separated by remaining vegetation strips to comply with legislative regulations. A maximum of 10 selected free-standing trees are left on the harvested area theses as part of the agreement with the nature protection authorities to preserve the species stability of the damaged area (Miklín and Čížek, 2014). In addition to these trees, all economically high-quality wood is removed during the clearcutting process. Furthermore, stumps were removed from the harvested area by a surface cutter, and the surface was prepared for the planting of seedlings. Typically, English oak is used for planting due to economic reasons (height of seedlings 30–50 cm). In the AOIs, the oak was planted at a stand density of around 9,100 seedlings per hectare (0.65 m × 1.70 m). Tree planting takes place according to legislative forestry standards.

2.2. METRIC model

METRIC, described by Allen et al. (2007a), is a diagnostic remote sensing model that estimates ET_a on a spatial scale based on surface energy balance calculations and satellite land surface temperature (LST) measurements. The model was designed for satellite images at a high spatial resolution, obtained from instance from the Landsat mission. In this study, data from both OLI (Operational land Imager) and TIRS (Thermal Infrared Sensor) sensors onboard Landsat 8 (L8) satellite were used to estimate ET_a at a 30-meter resolution.

2.2.1. Acquisition and preparation of remote sensing data

L8 satellite data are available on the USGS (United States Geological

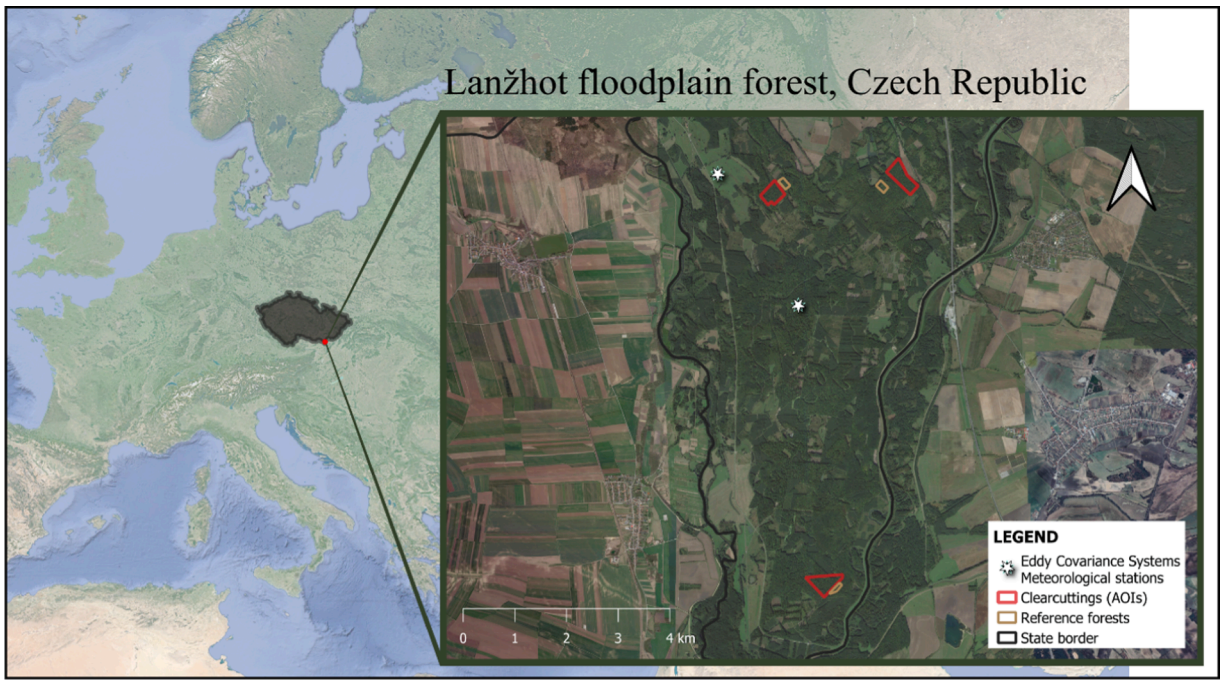


Fig. 1. Location of Lanžhot flood plain forest area in the southernmost tip of Moravia in the Czech Republic.

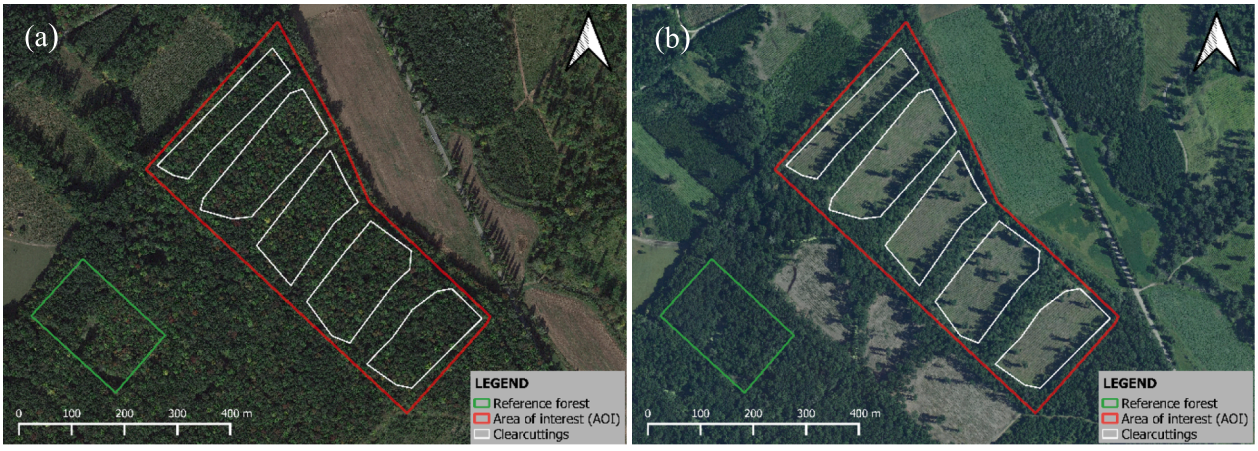


Fig. 2. a, b AOI 1 (red polygon) with clearcuttings (white polygons) and reference forest stand 1 (green polygon). Aerial photography from 2015 (a) and 2021 (b). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Survey) and NASA (National Aeronautics and Space Administration) websites (https://earthexplorer.usgs.gov/) (U.S. Geological Survey, 2022d). For this case study, the following datasets were used:

A) Surface Temperature data from L8 TIRS sensor, retrieved from Landsat Collection 1 Level-1 Science Products (U.S. Geological Survey, 2022a).
B) Surface reflectance data from L8 OLI sensor generated by EROS (Earth Resources Observation and Science) center, retrieved from Landsat Collection 2 Level-2 Science Products (U.S. Geological Survey, 2022b).
C) Data from the Digital Terrain Model of the SRTM1 sensor (Shuttle Radar Topography Mission), which was attached to the space shuttle Endeavor in 2001 (U.S. Geological Survey, 2022c).

The available satellite LST and surface reflectance data of the L8 satellite were selected according to the occurrence of cloud cover over the AOI. If there was cloud cover over one of the sites of interest in the Lanžhot area, this image was excluded from the available time series 2015–2021. The 66 clear sky images are available in the Lanžhot localities, with 52 available during the growing season (1. April – 30. October).

As described in Allen et al. (2007a), METRIC is based on the principle of detecting ET_a anchor pixels - "hot and cold pixels", where:

- **A hot pixel** is an area with a high surface temperature value and a low LAI value where ET_a reaches the lowest but not necessarily zero ET_a value in the AOI.
- **A cold pixel** is an area with low surface temperature and high LAI values where ET_a reaches the highest values in the AOI.

A simple soil water balance model was applied in the METRIC model to estimate soil water content of the hot pixel according to the FAO-56 and with that associated soil evaporation (Allen et al., 1998). The water balance model uses continuous time series of the meteorological data to compute ET_o and precipitation. These data were provided in 30

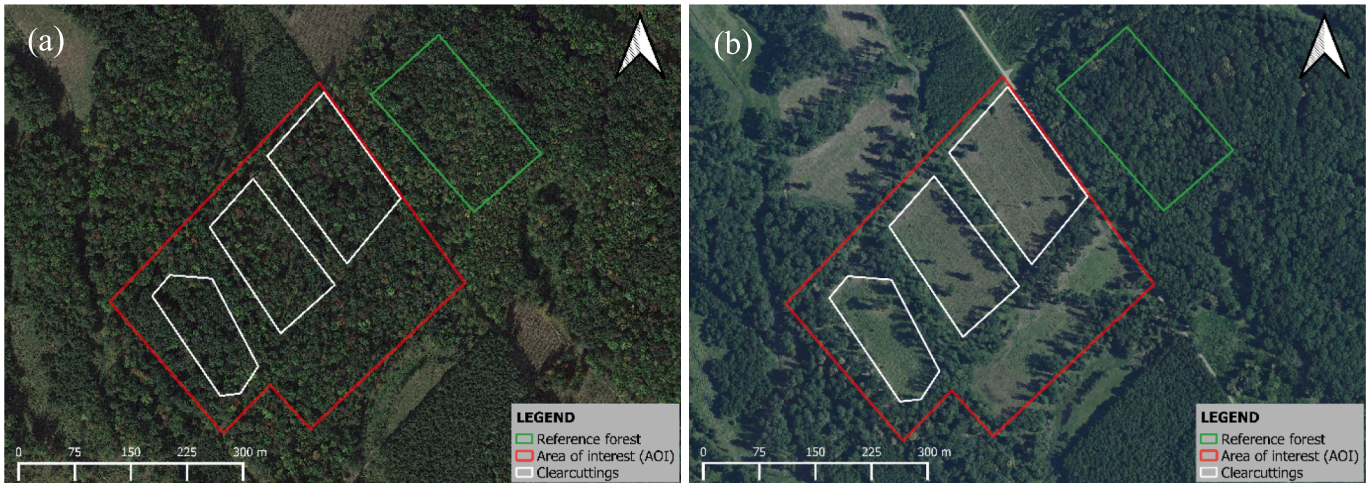


Fig. 3. a, b AOI 2 (red polygon) with clearcuttings (white polygons) and reference forest stand 2 (green polygon). Aerial photography from 2015 (a) and 2021 (b). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

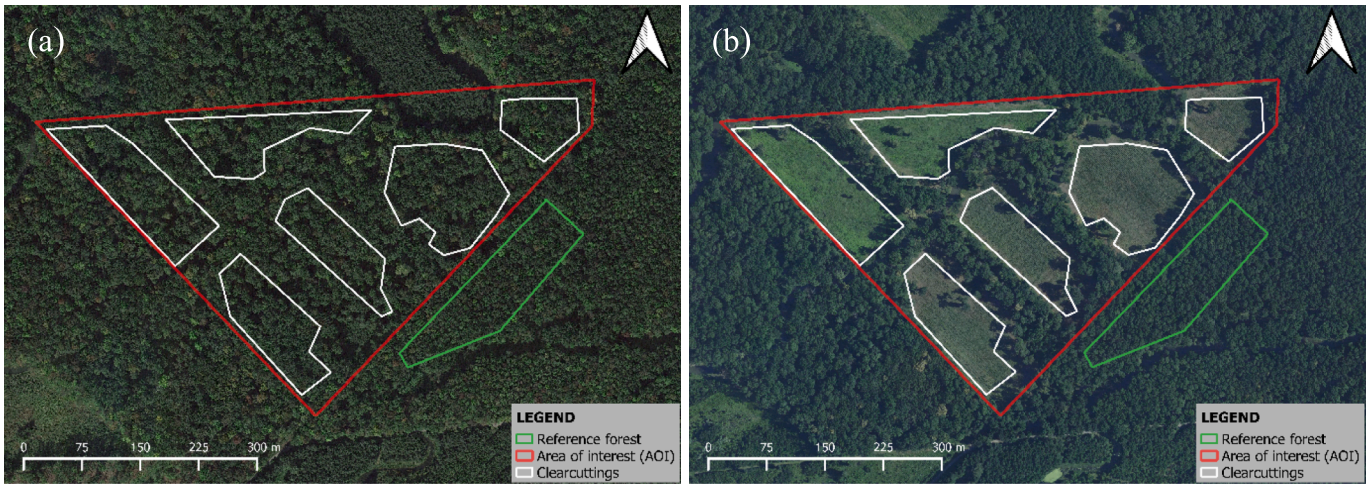


Fig. 4. a, b AOI 3 (red polygon) with clearcuttings (white polygons) and reference forest stand 3 (green polygon). Aerial photography from 2015 (a) and 2021 (b). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

min. time step from two meteorological stations, Lanžhot-Grassland and Lanžhot-Forest. Soil water balance was calculated for the topsoil defined as the uppermost 10 cm of soil that is typically considered as the most relevant for soil evaporation (Allen et al., 2005).

2.2.2. Calculation of LAI

LAI is calculated using surface reflectance data as suggested by Kang et al. (2016).

$$\mathrm{LAI} = \left(2.92\sqrt{\mathrm{EVI2}} - 0.43\right)^2$$

where EVI2 is the two-band enhanced vegetation index (Jiang et al., 2008). The EVI2 is calculated as.

$$\mathrm{EVI2} = \frac{2.5(\mathrm{NIR}_{\mathrm{band5}} - \mathrm{RED}_{\mathrm{band4}})}{(\mathrm{NIR}_{\mathrm{band5}} + 2.4\mathrm{RED}_{\mathrm{band4}} + 1)}$$

where NIR and RED are the reflectances for near infrared (band 5) and red (band 4), respectively, from L8 OLI.

2.3. Meteorological and EC data

Meteorological and EC data were obtained from the two ecosystem stations, Lanžhot-Grassland (measurement height 2 m) and Lanžhot-Forest (measurement height 44 m). The meteorological data were acquired at a half-hourly step. METRIC uses the following meteorological data: global radiation (W m^{-2}) (CNR 4, Kipp&Zonen, NE) and temperature (°C) ([1]. from 2015 to 01-01 to 2019–06-19 - EMS33, EMS Brno, CZ, [2]. from 2019 to 06-19 - HMP155, Vaisala, FI), half-hourly precipitation totals (mm 30 $min.^{-1}$) ([1] Thies Laser Precipitation Monitor, Adolf Thies, DE; [2] Weight Rain Gauge), relative humidity (%) (EMS33, EMS Brno, CZ; HMP155, Vaisala, FI), wind speed (m s^{-1}) (Wind Sonic 2D, Gill Instruments, UK).

METRIC requires meteorological data only for the day at the satellite overpass over AOI. However, because the METRIC uses the simple soil water balance model (Allen et al., 1998) to account for non-zero ET_a of hot pixel, the input meteorological data from meteorological stations were entered into the model as continuous-time series at a half-hourly time step for a period with continuous observation (for Lanžhot-Forest it is 2015–2021, for Lanžhot-Grassland it is 2019–2021).

The ET_o was calculated by the Penman-Monteith method specified in the FAO-56 methodology (Allen et al., 1998). The input meteorological data were acquired from the meteorological station Lanžhot-Forest. In this case, the ET_o represents the evapotranspiration values of a hypothetical reference grass surface with ideal growth conditions and is not limited by a lack of water. In this study, ET_o was used to determine the ET_a/ET_o, which better represents the ET_a rate estimated by the METRIC

model under given meteorological conditions.

The available ET_a ground measurement data from the EC system were derived from the latent heat flux measurements. Two different ecosystems EC sites (Lanžhot-Forest, Lanžhot-Grassland) were used to validate METRIC in the Lanžhot area. The first EC station is located in the floodplain forest (Fig. 5 a, c). The second EC station is located in the grassland (Fig. 5 b, d) at a distance of 3 km from the first EC station. Both EC systems used in the study consisted of an enclosed path CO_2/H_2O gas analyzer Li-7200 (LI-COR Environmental, Lincoln, USA) and a 3D sonic anemometer Gill HS-50 (Gill Instruments Limited, Hampshire, UK). The raw data were stored in the Li-7550 Analyzer Interface Unit (LI-COR Environmental, Lincoln, USA) at a sampling frequency of 20 Hz. The measurement height of EC for the Lanžhot-Grassland station was 2 m above the soil surface. The measurement height of EC varied between 44 m and 48 m during the measurement period for the Lanžhot-Forest (44 m = 2015–01-01 to 2018–12-05; 48 m = 2018–12-05 to 2021–10-19; 44 m = 2021–10-19 to 2021–12-31). The standard flux averaging interval was 30 min. Values for longer periods were derived via aggregation of gap-filled 30-minute values. The latent heat flux data (W m^{-2} or MJ m^{-2} day^{-1}) were converted to mm day^{-1} according to Allen et al. (1998). The EC data were used for the evaluation of METRIC.

2.4. METRIC model evaluation by EC ground measurements

The simple two-dimensional parameterization method for flux footprint prediction (Kljun et al., 2015) was applied to estimate the daily mean EC footprint for both stations. The daily mean footprint was determined as an average of footprints during the period representing 80% of the daily cumulative ET_a. In this way, the large nighttime footprints (stable atmospheric stratification) are considered irrelevant for the evaluation. These mean daily footprints were applied to crop the METRIC ET_a and compute a spatially weighted mean of ET_a from METRIC for comparison with daily ET_a from EC.

The energy balance closure was computed for both EC stations in Lanžhot. The sensible heat flux (H, W m^{-2}) and latent heat flux (W m^{-2}) were measured and computed using EC systems and tools (Section 2.5) at the same measurement height. The net (total) radiation (W m^{-2}) (CNR 4, Kipp & Zonen, NE) was measured at 44 m height. The soil heat flux (W m^{-2}) was determined from the measurements (HFP01SC-05, Hukseflux, NE) in a 5 cm depth.

The METRIC model was evaluated using clear sky L8 images in 2015–2021. METRIC ET_a outputs were compared to two ET_a measurements of EC measurements at two independent stations in the Lanžhot area (Lanžhot-Forest, Lanžhot-Grassland). In this study, 66 L8 clear sky satellite images were available at the Lanžhot-Forest station during the whole period 2015–2021, and 28 clear sky images were available at the Lanžhot-Grassland station between.

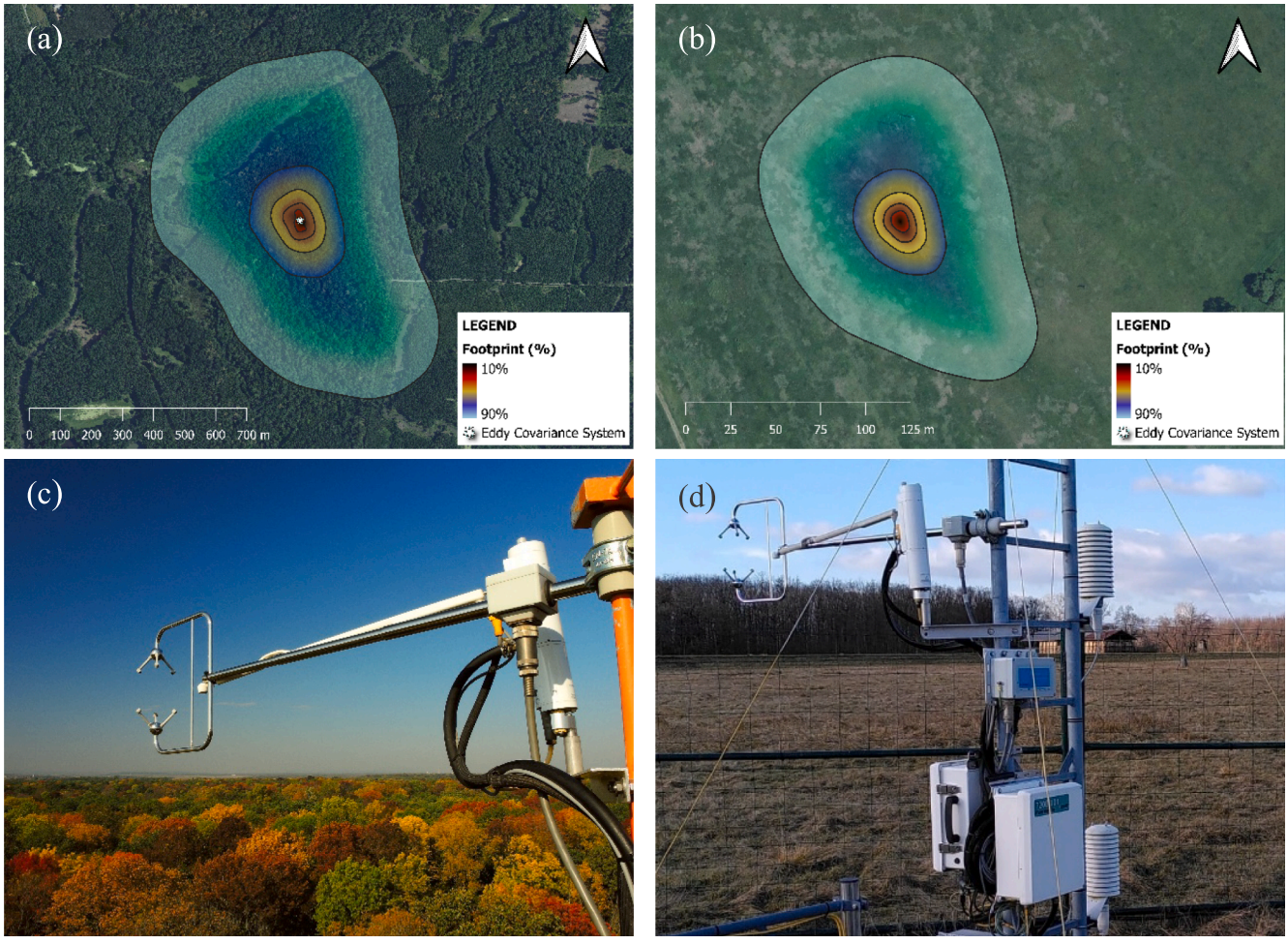


Fig. 5. Footprints of the eddy covariance stations in Lanžhot forest (a) and grassland (b) sites. The polygons (black lines) represent cumulative contribution to measured ET_a starting from 10% around the tower (dark red color), through 30%, 50%, 70%, up to 90% polygon (light blue color). Bottom photographs illustrate the eddy covariance system in the forest (c) and grassland (d). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2.5. Evapotranspiration and LAI detection of the clearcut area

ET_a, ET_a/ET_o, ET_a/ET_o/LAI (normalized evapotranspiration-leaf area efficiency), and LAI were modeled by METRIC in the growing season in 2015–2021 because the estimation of ET_a in the growing season allows quantification of the ET_a rate changes in the AOI areas during removal and reforestation. The 52 clear sky L8 images were available for the detection of ET_a by METRIC in growing seasons between 2015 and 2021 in AOI 1–3. ET_a, ET_a/ET_o and LAI outputs of the clearcuttings were also compared to the average values of selected reference forest areas (Figs. 3, 4, 5) which were not harvested (Section 3.3-3.5).

ET_a, ET_a/ET_o and LAI map outputs were selected for specific days in 2015–2021 according to the following criteria:

1. Selected images were taken during the summer period.
2. The evaporation coefficient (ET_a/ET_o) for the hot pixels in METRIC had to be lower than 0.4.

The first condition provides similarity in terms of the phenological phases of vegetation. The second condition ensures a similar and rather low contribution of soil evaporation to ET_a in the images. The second conditions also assume that ET_a of the hot pixel (if not zero) is dominated by soil evaporation rather than transpiration due to low LAI of that pixel.

2.6. Statistical analyses

Data analyses were performed using R (R Core Team, 2022) statistical programming language in RStudio (RStudio Team, 2020). The data were processed by analysis of variance (ANOVA) with a linear mixed effect (lme) model with denominator degrees of freedom calculated according to Pinheiro and Bates (2000). The response variables included ET_a, ET_a/ET_o, LAI and ET_a/ET_o/LAI. Year was considered as the fixed effect and AOI, and day of the year as the random effect. An autocorrelation function structure for the linear mixed effects was applied to account for the dependence of the consecutive measurements repeated along the year. Tukey's honestly significant difference post-hoc test with Westfall's adjustment for multiplicity (Bretz et al., 2010; Westfall, 1997) was subsequently applied on the mixed effects linear model to identify statistically significant differences between the years. Assumptions on distributions of residuals of the linear models were checked by visual inspection of quantile–quantile probability plots and by analysis of skewness and kurtosis. Since the normal distribution of residuals was not detected, the original values were transformed according to the Box-Cox procedure (Venables and Ripley, 2002). The differences were considered significant when $P \leq 0.05$.

3. Results

3.1. Evaluation of the METRIC model

METRIC was evaluated using two ground point measurements of EC stations within two different biotopes - Lanžhot-Forest and Lanžhot-Grassland (Fig. 6 a, b). The EC energy balance closure was 0.89 for the Lanžhot-Forest station and 0.84 for the Lanžhot-Grassland station in 2015–2021. Evaluation shows relatively good agreement between METRIC ET_a and EC measurements at the Lanžhot-Forest station (coefficient of determination (R^2) = 0.86, root-mean-square error (RMSE) = 0.88 mm day^{-1}, mean absolute error (MAE) = 0.69 mm day^{-1}, Fig. 6 a). In average, METRIC overestimated ET_a in the forest biome, with a mean error (ME) of 0.61 mm day^{-1}. The good agreement of measurements between METRIC model outputs and EC is supported by the results of the second evaluation in Lanžhot-Grassland (Fig. 6 b) (R^2 = 0.73, RMSE = 0.70 mm day^{-1}, MAE = 0.58 mm day^{-1}). Also in the case of grassland the model METRIC in average overestimated the EC ET_a measurements (ME = 0.21 mm day^{-1}). The METRIC model evaluated in this way was deemed to be sufficiently reliable for assessment the impacts of clearcutting on ET_a in the three AOIs.

3.2. LAI and ET_a results

The ET_a and LAI outputs (Figs. 7 to 9) show the annual time series of ET_a, ET_a/ET_o and LAI images acquired on selected summer days in the 2015–2021 period. The summer images of 2017 (Figs. 7-9) show a decrease in ET_a, ET_a/ET_o and LAI in AOIs 1–3 after forest logging. In this growing season, METRIC detected an annual decrease in the ET_a/ET_o by 35–39% on average (ET_a by 35–40%), while LAI annual values decreased by 70–80% on average compared to the average annual values in AOIs 1–3 of 2015–2016. The LAI values were lower than LAI = 1 in all AOIs 1–3 (LAI = 0.72, 0.69; 0.81) in these years, which indicates

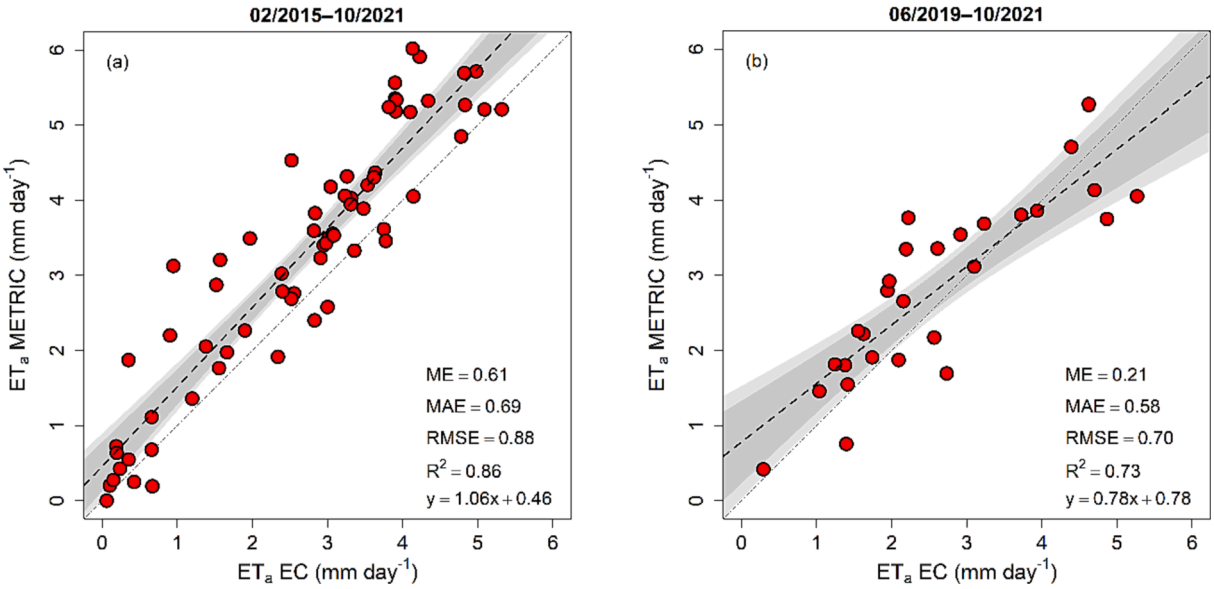


Fig. 6. a, b Scatter plots of the comparison of ET_a METRIC to surface measurement ET_a EC in Lanžhot-Forest (a) and Lanžhot-Grassland (b). The dashed black line shows the trend of validated values. The grey thin chain line displays the ideal hypothetical match (1:1) between methods. The plots are supplemented by confidence intervals at a confidence level of 95% (inner grey belts) and 99% (outer light grey belts).

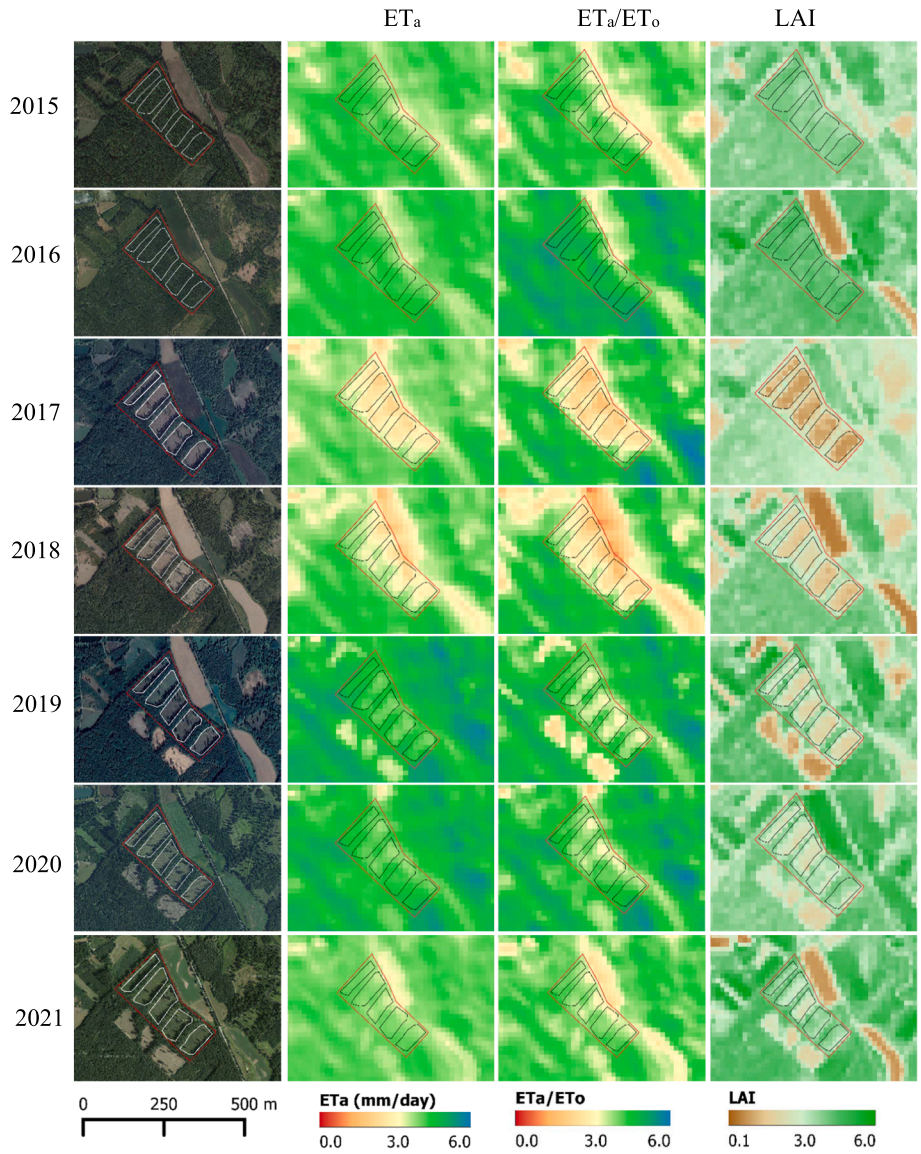


Fig. 7. Selected ET_a, ET_a/ET_o and LAI summer images in the period 2015–2021 of AOI 1. The red line is the border of AOI 1. The black or white line is the border of one clearcutting area. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

a surface without significant tree vegetation cover.

The lower values of the ET_a, ET_a/ET_o, and LAI agree with the method of removal of the vegetation cover and the subsequent technical processing and modification of the AOI surfaces when the vegetation cover was completely removed, and the soil surface was mechanically modified before the planting of the new generation of trees. The clearcutting areas are separated by vegetation belts, which are left there following the legislative rules. These vegetation belts are visible between clearcutting areas, where the LAI value is greater than the LAI value in the clearcutting areas. In the dry growing season of 2018, the consequences of the removal of vegetation became apparent again. This year, the ET_a/ET_o was up to 21% lower, ET_a was lower by up to 37% and LAI was as low as 79% than in the 2015 and 2016 seasons on average.

In 2019 and 2020, the average ET_a/ET_o of AOI 1–3 reached 101% and 104%, respectively, compared to 2015–2016. While the ET_a values were still lower by approximately 20% and LAI by 50–40% in the same period compared to the 2015 and 2016 averages, the ET_a/ET_o values reached the level as before clearing the forest. The ET_a/ET_o values from the last three years indicate a recovery of the ET_a/ET_o after three years of forest disturbance and indicate that the pioneer species (Betula pendula, Rubus spp) with planted trees (English oak) entirely reached the ET_a and ET_a/ET_o of the mature forest in the floodplain forest habitat. The partial recovery of LAI was recorded in the last year of monitoring. The LAI values were approximately 27% and 6% lower on average in the first and second clearcuttings, respectively, while in the third clearcutting, the average LAI values were similar compared to the periods before targeted logging.

3.3. Seasonal course of ET_a and ET_a/ET_o in 2015–2021

Fig. 10 shows the ET_a seasonal course in the period 2015–2021. The estimated ET_a outputs from METRIC model show the results of the specific days when the satellite images were acquired. The decrease of the ET_a compared to the reference forest (difference by 22% on average) and previous years 2015 and 2016 (-30%) is evident from the plot due to the removal of vegetation cover. According to the method of logging and technical preparation of the surface, the level of ET_a in 2017 and 2018 should be mostly represented by soil evaporation because transpiration should be strongly reduced due to the removal of vegetation cover. The ET_a decline after logging is also confirmed by the ANOVA post-hoc test indicating that ET_a in 2017 and 2018 is significantly lower as compared to ET_a in 2015 and 2016 (Fig. 10). The highest values of ET_a for AOIs and

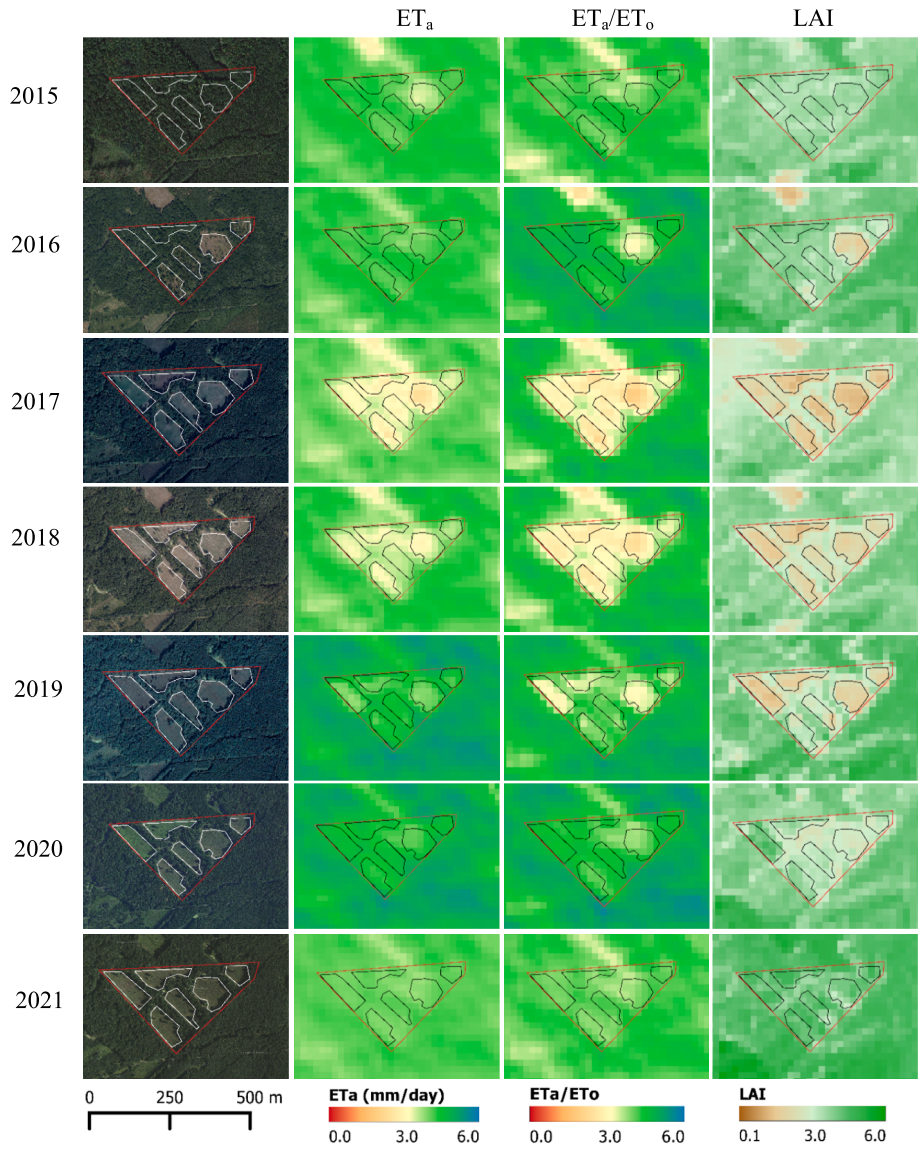


Fig. 8. Selected ET_a, ET_a/ET_o and LAI summer images in the 2015–2021 period of AOI 2. The red line is the border of AOI 2. The black or white line is the border of one clearcutting area. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

reference forests were measured in the summer of 2021, when ET_a reached almost 7 mm day^{-1}. In this growing season, the similar daily values ET_a and in the AOIs a reference forest were detected. This was also supported by the ANOVA post-hoc test, which indicated that ET_a in 2020 and 2021 was not significantly different to ET_a in the pre logging years 2015 and 2016 (Fig. 10).

A decrease in the ET_a/ET_o was recorded in all AOIs compared to reference forests in the growing season in 2017 after the removal of vegetation cover in 2016. ET_a/ET_o decreased by 13% on average compared to the reference forest in the growing season of 2017. The lower values compared to reference were also recorded in the 2018 and 2019 growing seasons (for ET_a/ET_o by 12% and 6% on averages). The ANOVA post-hoc test indicated that ET_a/ET_o in 2017 and 2018 were significantly lower compared to pre logging years 2015 and 2016 (Fig. 11). The full recovery of ET_a/ET_o was recorded in the fourth growing season after the removal of vegetation cover when ET_a/ET_o almost reached the values (2020 = 97%, 2021 = 99% on average) of the reference forest. Based on the ANOVA pos-hoc test, the recovery of ET_a/ET_o can be identified even one year earlier, i.e. in 2019 when the ET_a/ET_o was not already significantly different from the values in pre logging years 2015 and 2016 (Fig. 11).

3.4. Seasonal course of LAI in 2015–2021

A significant decrease in LAI was recorded in all AOIs compared to reference forests in the growing season in 2017 after the logging of the forest (Fig. 12). The LAI decreased by 50% on average compared to the reference forest that year. The significant change of LAI was also identified by results of ANOVA post-hoc test suggesting that LAI during period 2017–2020 was significantly different form the values in pre logging years 2015 and 2016 (Fig. 12).

Lower LAI values compared to the reference forest were also recorded in the 2018 and 2019 growing seasons (for LAI by 31% and 23% in average). While the full recovery of ET_a/ET_o was recorded in 2020 this year LAI was lower by 13% on average compared to the reference stand. The recovery of LAI was detected a year later, in 2021 when LAI values reached 98% on average compared to the reference forest. The ANOVA post-hoc test also shows and confirms the slower recovery of LAI compared to ET_a/ET_o (cfr. Fig. 11 and Fig. 12).

3.5. Course of ET_a/ET_o/LAI ratio in 2015–2021

Fig. 13 shows the progress of ET_a/ET_o/LAI in the period of

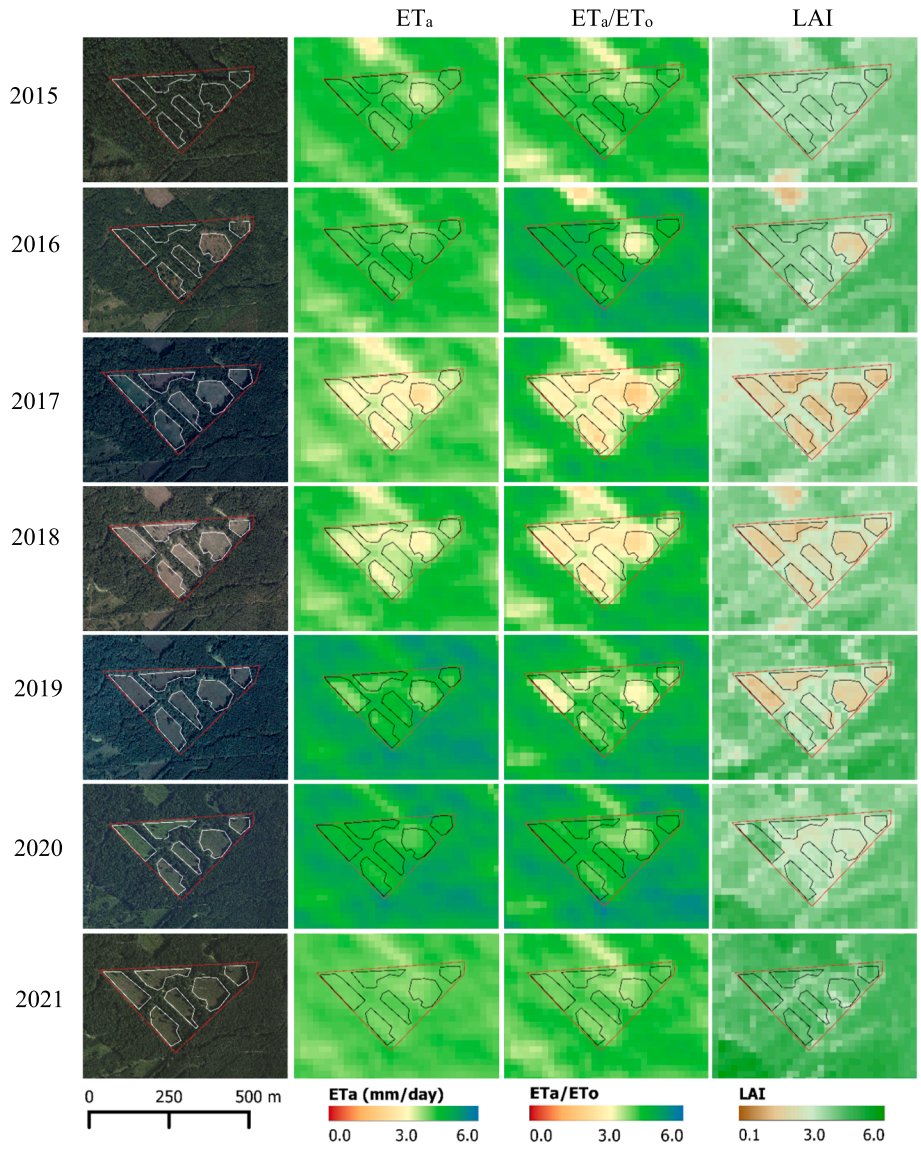


Fig. 9. Selected ET_a, ET_a/ET_o and LAI summer images in the 2015–2021 period of AOI 3. The red line is the border of AOI 3. The black or white line is the border of one clearcutting area. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2015–2021. ET_a/ET_o/LAI expresses the normalized (by ET_o) efficiency of ET_a per unit of LAI. The significant increase of ET_a/ET_o/LAI compared to the reference forest was recorded in 2017 due to vegetation removal by targeted logging, which caused a higher decrease in LAI than ET_a/ET_o. Increased values of the efficiency were also recorded in 2018. In the last two years, the values of ET_a/ET_o/LAI were similar to the reference forest stand values. The mean ET_a/ET_o/LAI values of the reference forest stand or AOIs before logging ranged fluctuated around 0.3 while after logging the values were around 0.8 and 0.6 in 2017 and 2018, respectively.

4. Discussion

The evaluation of METRIC ET_a outputs confirmed a good agreement with EC measurement at two different stations Lanžhot-Forest and Lanžhot-Grassland. The comparison showed that METRIC outputs slightly overestimated the ET_a outputs compared to ground point measurements. The overestimation can be partly explained by lack of energy balance closure in EC measurements (Lanžhot-Forest = 0.86; Lanžhot-Grassland = 0.73) but also various types of assumptions and uncertainties in METRIC model including correct determination of ET_a at hot (dependent on soil water balance modelling) and cold pixels (dependent on the assumption about potential evapotranspiration), as well as challenges related to scaling an instantaneous latent heat flux to daily ET_a (Allen et al., 2007a). The model evaluated in this way, can be further used in the testing of hypotheses aimed at the quantification of individual hydrological processes in the landscape. The subject of future studies can be placed not only on the production potential of agro-ecosystems and forest stands but also on the ecological solution of the biotically valuable area of the floodplain forest ecosystem, which, due to the recent partial drainage of the area, is more susceptible to fluctuations in the groundwater level, as explained in the study by Čermák and Practice (2009). An advantage of using remote sensing-based ET_a models, such as METRIC, is mapping ET_a at the high spatial resolution, in orders of tens of meters, which is potentially suitable for the heterogeneous landscape of Central Europe within the framework of local studies and analyses.

Our results show the reduction of ET_a, ET_a/ET_o, and LAI after the almost complete removal of vegetation cover (except 10 trees per hectare). The results correspond with other studies that demonstrated a similar effect of ETa/ETo and LAI reduction on hydrological processes (Kubota et al., 2021; Sun et al., 2017), which affects the underground

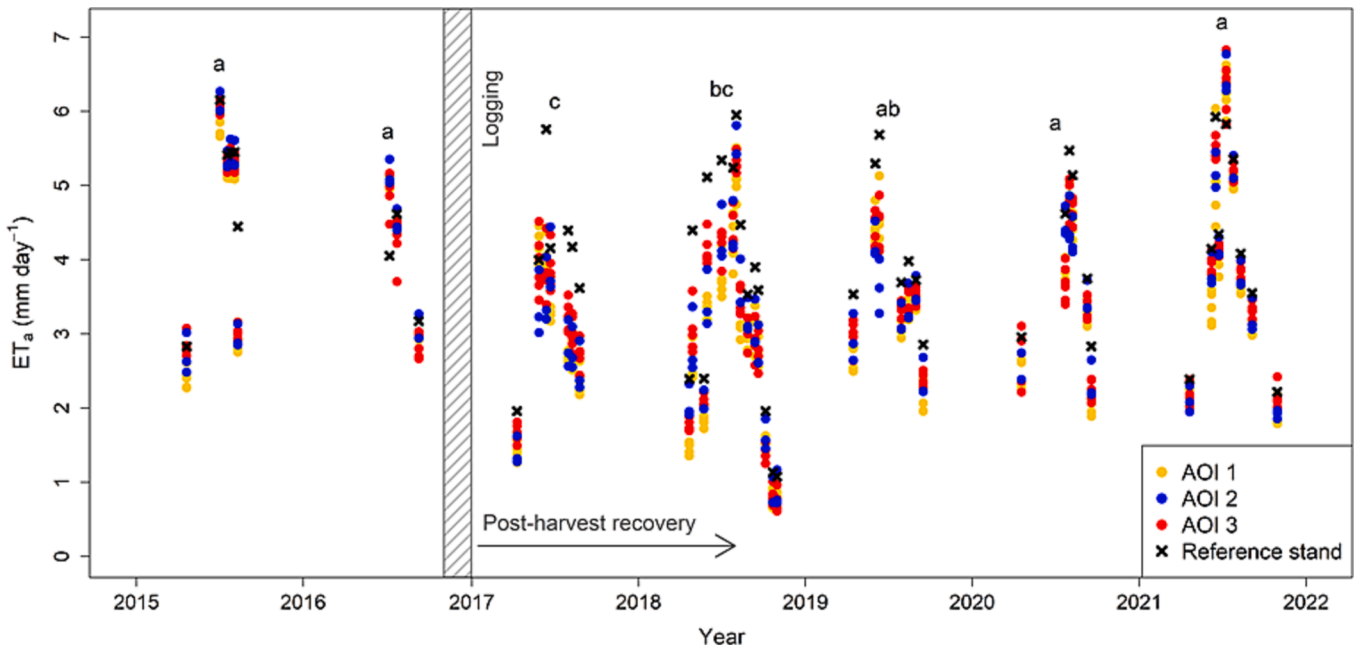


Fig. 10. ET_a temporal variation in AOIs 1–3 and reference forest in the growing season 2015–2021. The years with the same letters are not statically significantly different ($P > 0.05$).

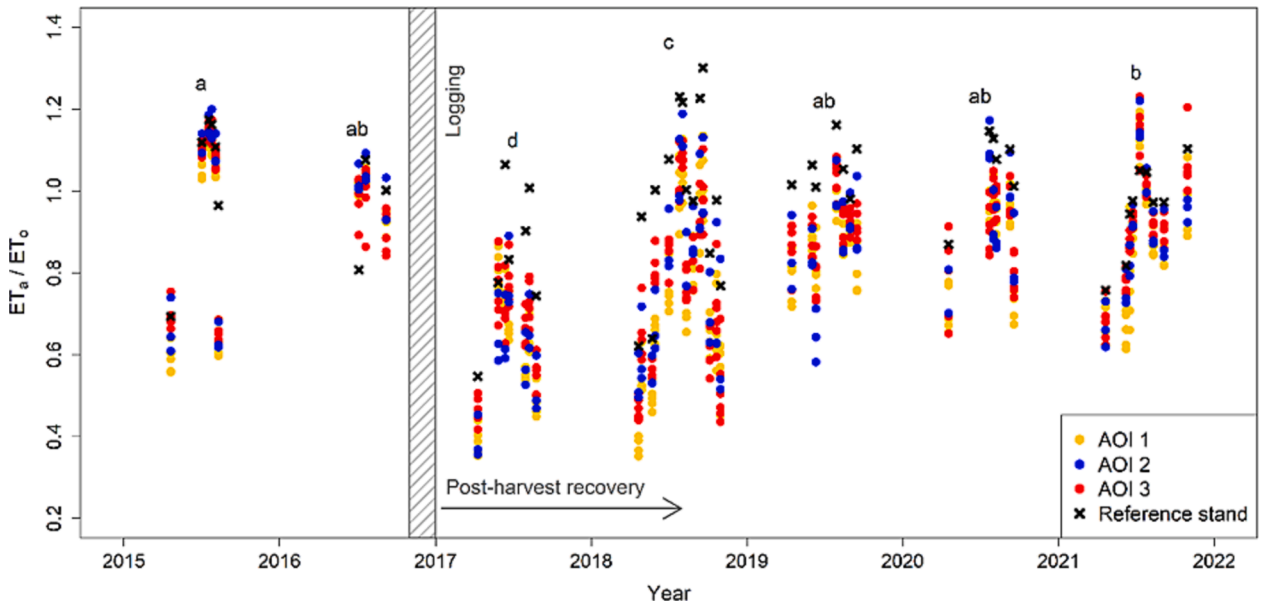


Fig. 11. The ET_a/ET_o temporal variation in AOIs 1–3 and reference forest in the growing season 2015–2021. The years with the same letters are not statically significantly different ($P > 0.05$).

water level (Stasik et al., 2020), interception (Sun et al., 2017) or runoff (Dung et al., 2012; Chiu et al., 2022; Kuraji et al., 2019; Yang, 2019; Thompson et al., 2018). While LAI in our study decreased by 80%, the ET_a/ET_o dropped by 40% after logging. The results show that the regeneration of the ET_a rate and ET_a/ETo occurred faster than the LAI in the floodplain forest. According to the mean values before and after forest logging and comparison with the reference forest, the recovery of ET_a/ET_o was detected one year earlier than a recovery of LAI. This was confirmed by the statistical results of the post-hoc ANOVA test, which show a recovery of ET_a/ET_o even two years earlier than the comparison between the reference forest and the averages of the growing season data. The faster recovery of ET_a as compared to LAI can be explained by the relationship between ET_a and vegetation parameters. In general, LAI development is closely related to ET_a and to the available energy partitioning (Iritz and Lindroth 1996; Zenone et al., 2015). The core of this dependence is a non-linear positive relationship between potential transpiration and LAI (Allen et al., 1998; Fischer et al., 2018). The maximum transpiration is typically approached for relatively low LAI values around 1.5–3 where denser canopy does not transpire significantly more water because of the energy limitations (Iritz and Lindroth, 1996; Blanken et al., 2001; Fischer et al., 2018). This energy limitation come naturally from the Beer-Lambert law providing that the radiation interception within a canopy is highly non-linearly related to LAI (Larcher, 2003). The near maximum transpiration around LAI of 1.5–3 supports also the results of this study where ET_a/ET_o/LAI was close to 0.3 for mature or recovered stands with LAI around and 4, and close to

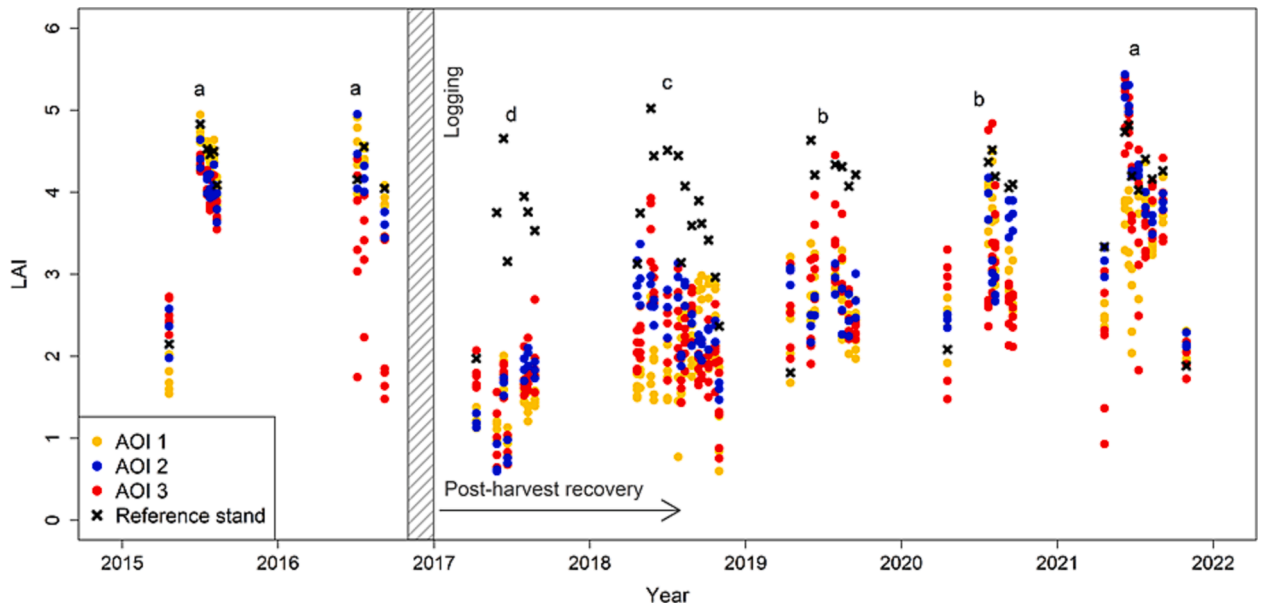


Fig. 12. The LAI temporal variation in AOIs 1–3 and reference forest in the growing season 2015–2021. The years with the same letters are not statically significantly different ($P > 0.05$).

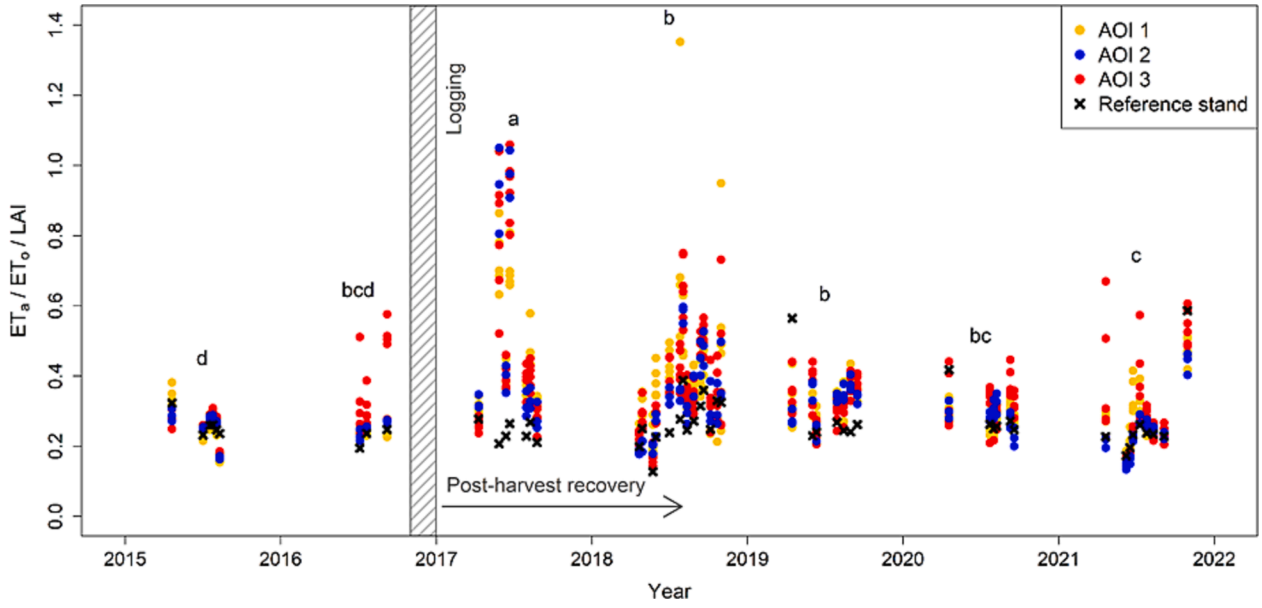


Fig. 13. The ET_a temporal variation in AOIs 1–3 and reference forest in the growing season 2015–2021. The years with the same letters are not statically significantly different ($P > 0.05$).

0.8 and 0.6 for post-harvest AOIs with LAI around 1.5 and 2 in 2017 and 2018, respectively.

Moreover, the regeneration of ET_a and ET_a/ET_o occurred much faster than in some other studied forest biomes, such as the forest biomes reported by Kubota et al. (2021), Yang et al. (2017), where recovery of ET_a was recorded after 20 years. This increase in regeneration may be due to the higher water availability in this AOI area. The full ET_a/ET_o recovery was recorded for the fourth growing season after vegetation cover removal, while the LAI recovery was recorded for the fifth growing season. The faster regeneration of ET_a and ET_a/ET_o compared to other study areas, as Kubota et al. (2021) can be caused by a higher groundwater level in the AOIs of this study. The higher groundwater level can be more accessible for new tree generation, resulting in faster coverage of the disturbed area with a new vegetation cover capable of quickly replacing the transpiration rate of the mature stand.

Čermák and Prax (2009), in their analysis of tree species of floodplain forests, confirmed the lower rate of transpiration of individual young English oak trees compared to older trees. However, when detecting the spatial distribution of ET_a in a given area, as shown in this study, it is necessary to look at the planting density of the reforestation, which is very important for the spatial distribution of ET_a. For example, according to forestry standards and procedures, up to 10,000 English oaks are planted per hectare. In addition, the vegetation cover of the trees planted in this way is completed with pioneering tree species and herbaceous nature, which further significantly increases the transpiration rate of the disturbed area and can thereby equalize the difference in ET_a between the mature stand before harvesting and the new generation of trees.

The results of this study are affected by the very dry and warm period 2015–2020 (Řehoř et al., 2020; Trnka et al., 2020; Zahradníček et al., 2021), especially in 2017 and 2018, when strong vegetation stress caused by drought was recorded in the late summer (Kowalska et al., 2020). Except for meteorological and pedological factors, the speed of recovery of the ET_a rate to the level before the logging of the stand also depends on the subsequent technical treatment of the surface, which usually precedes the planting of a new generation of trees and suppresses the process of natural succession of the AOI. The study of López et al. (2009) demonstrates that the response of holm oak (*Quercus ilex* L.) stands to consecutive disturbances highly depends on the intensity of the disturbance. Our work represents conditions where all vegetation cover is removed from the AOI, the bare soil surface is leveled, and dense planting of English oak seedlings follows. Study (Brychta et al., 2020) show that some loggings methods such as Shelterwood cutting may improve the efficiency of water management in stands and limit the often inappropriate interventions during its logging with regard to the biodiversity of the area (Roleček and Řepka, 2020), which is significantly suppressed in the current forestry procedures of extraction and restoration of forests in the floodplain areas (Miklín and Čížek, 2014).

5. Conclusions

In this study, the METRIC model was successfully evaluated in Central European conditions. The results show good agreement between EC measurements and METRIC ET_a outputs on two different vegetation cover surfaces, floodplain forest and grassland, and confirmed that METRIC can be reused in similar studies applied to different vegetation surfaces in the area. The evaluated model was used to test the hypotheses. According to Kubota et al. (2018) we hypothesized a significant decrease in the rate of ET_a and LAI is predicted in response to the removal of vegetation cover. In addition, we assumed that recovery of the ET_a, ET_a/ET_o, and LAI on the pre logging values is a long term process on the order of decades due to vegetation changes that affect the local climatic conditions. of interest. The results showed a significant reduction in the amount of LAI and ET_a/ET_o due to the targeted removal of vegetation cover and soil preparation before the planting of a new generation of trees. Furthermore, ET_a/ET_o recovery was noted in the fourth year after targeted logging and was one year faster than LAI recovery, indicating that the ET_a rate is not directly dependent on vegetation cover density. The results also showed that ET_a/ET_o recovery in floodplain forest biomes is faster than in other types of forest biomes reported in earlier studies, which may be due to higher water availability. The rapid recovery of the ET_a/ET_o suggests that the impacts of logging on the hydrology may be temporally limited in biomes such as floodplain forests. This can help to restore the pre logging ecosystem conditions and species diversity more quickly in such biologically valuable areas.

CRediT authorship contribution statement

T. Ghisi: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Funding acquisition, Visualization, Writing - original draft, Writing - review & editing. **M. Fischer:** Conceptualization, Formal analysis, Methodology, Funding acquisition, Supervision, Writing - original draft, Writing - review & editing. **N. Kowalska:** Validation, Writing - original draft, Writing - review & editing. **G. Jocher:** Validation, Writing - original draft, Writing - review & editing. **M. Orság:** Data curation, Writing - original draft. **M. Bláhová:** Formal analysis, Writing - original draft. **H. Nieto:** Formal analysis, Writing - review & editing. **L. Homolová:** Methodology, Data curation, Writing - review & editing. **Z. Žalud:** Supervision, Project administration, Funding acquisition, Writing - review & editing. **M. Trnka:** Conceptualization, Supervision, Project administration, Funding acquisition, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The research infrastructure and CzechGlobe team was financially supported by the project SustES - Adaptation strategies for sustainable ecosystem services and food security under adverse environmental conditions (CZ.02.1.01/0.0/0.0/16_019/0000797). M.F. was supported by project SS01010207 of the Technology Agency of the Czech Republic entitled “The development of tool for identification of the main water management risks at the Thaya river basin and the methodology addressing their systematic solution in the conditions of climate change”. M.T. was supported by project SS02030040 of the Technology Agency of the Czech Republic PERUN – “Prediction, Evaluation and Research for Understanding National sensitivity and impacts of drought and climate change for Czechia”. Work of T.G. was also financially supported by the CzechGlobe grant agency (CzeGGA) – ” Validation of actual evapotranspiration by model METRIC in the Czech landscape”.

References

Allen, R.G., Pereira, L.S., Raes, D., Smith, M. (1998). Crop evapotranspiration-Guidelines for computing crop water requirements-FAO Irrigation and drainage paper 56. Fao, Rome, 300(9), D05109, available at: https://www.researchgate.net/publication/235704197_Crop_evapotranspiration-Guidelines_for_computing_crop_water_requirements-FAO_Irrigation_and_drainage_paper_56 (last access: 11 September 2022).

Allen, R.G., Pruitt, W.O., Raes, D., Smith, M., Pereira, L.S., 2005. Estimating evaporation from bare soil and the crop coefficient for the initial period using common soils information. J. Irrig. Drain. Eng. 131 (1), 14–23. https://doi.org/10.1061/(ASCE)0733-9437(2005)131:1(14).

Allen, R.G., Tasumi, M., Trezza, R., 2007a. Satellite-based energy balance for mapping evapotranspiration with internalized calibration (METRIC)—Model. J. Irrig. Drain. Eng. 133 (4), 380–394. https://doi.org/10.1061/(ASCE)0733-9437(2007)133:4(380).

Allen, R.G., Tasumi, M., Morse, A., Trezza, R., Wright, J.L., Bastiaanssen, W., Robison, C. W., 2007b. Satellite-based energy balance for mapping evapotranspiration with internalized calibration (METRIC)—Applications. J. Irrig. Drain. Eng. 133 (4), 395–406. https://doi.org/10.1061/(ASCE)0733-9437(2007)133:4(395).

Anderson, M.C., Norman, J.M., Diak, G.R., Kustas, W.P., Mecikalski, J.R., 1997. A two-source time-integrated model for estimating surface fluxes using thermal infrared remote sensing. Remote Sens. Environ. 60 (2), 195–216. https://doi.org/10.1016/S0034-4257(96)00215-5.

Bastiaanssen, W.G., Menenti, M., Feddes, R.A., Holtslag, A.A.M., 1998. A remote sensing surface energy balance algorithm for land (SEBAL). 1. Formulation. Journal of hydrology, 212, 198-212, available at: https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=6137f8f42b7662ddd1add19064f99f5ffe44b99d (last access: 11 September 2022).

Beudert, B., Bernsteinová, J., Premier, J., Bässler, C. (2018). Natural disturbance by bark beetle offsets climate change effects on streamflow in headwater catchments of the Bohemian Forest. Silva Gabreta, 24, 21-45, available at: https://www.npsumava.cz/wp-content/uploads/2019/06/2_sg_24_beudertetal.pdf (last access: 11 September 2022).

Blanken, P.D., Black, T.A., Neumann, H.H., Den Hartog, G., Yang, P.C., Nesic, Z., Lee, X., 2001. The seasonal water and energy exchange above and within a boreal aspen forest. J. Hydrol. 245 (1–4), 118–136. https://doi.org/10.1016/S0022-1694(01)00343-2.

Bretz, F., Hothorn, T., Westfall, P., 2010. Multiple Comparisons Using R, 1st ed. Chapman and Hall/CRC. https://doi.org/10.1201/9781420010909.

Brychta, J., Bílek, L., Linda, R., Vítámvás, J., 2020. Does shelterwood regeneration on natural Scots pine sites under changing environmental conditions represent a viable alternative to traditional clear-cut management? Central Eur. Forestry J. 66 (2), 104–115. https://doi.org/10.2478/forj-2020-0014.

Carrasco-Benavides, M., Ortega-Farías, S., Lagos, L.O., Kleissl, J., Morales-Salinas, L., Kilic, A., 2014. Parameterization of the satellite-based model (METRIC) for the estimation of instantaneous surface energy balance components over a drip-irrigated vineyard. Remote Sens. (Basel) 6 (11), 11342–11371. https://doi.org/10.3390/rs61111342.

Chandel, A.K., Molaei, B., Khot, L.R., Peters, R.T., Stöckle, C.O., 2020. High resolution geospatial evapotranspiration mapping of irrigated field crops using multispectral and thermal infrared imagery with metric energy balance model. Drones 4 (3), 52. https://doi.org/10.3390/drones4030052.

Chiu, C.W., Gomi, T., Hiraoka, M., Shiraki, K., Onda, Y., Dung, B.X., 2022. Evaluating changes in catchment-scale evapotranspiration after 50% strip-thinning in a headwater catchment. Hydrol. Process. 36 (6), e14611.

Čermák, J., Prax, A., 2009. Transpiration and soil water supply in floodplain forests. Ekologia 28 (3), 248–254. https://doi.org/10.4149/ekol_2009_03_248.

Dung, B.X., Gomi, T., Miyata, S., Sidle, R.C., Kosugi, K., Onda, Y., 2012. Runoff responses to forest thinning at plot and catchment scales in a headwater catchment draining Japanese cypress forest. J. Hydrol. 444–445, 51–62. https://doi.org/10.1016/j.jhydrol.2012.03.040.

Fischer, M., Zenone, T., Trnka, M., Orság, M., Montagnani, L., Ward, E.J., Ceulemans, R., 2018. Water requirements of short rotation poplar coppice: Experimental and modelling analyses across Europe. Agric. For. Meteorol. 250, 343–360. https://doi.org/10.1016/j.agrformet.2017.12.079.

Hankerson, B., Kjaersgaard, J., Hay, C., 2012. Estimation of evapotranspiration from fields with and without cover crops using remote sensing and in situ methods. Remote Sens. (Basel) 4 (12), 3796–3812. https://doi.org/10.3390/rs4123796.

Hruška, J., Lamačová, A., & Chuman, T. (2016). Bezzásahový režim nemá zásadní vliv na hydrologii šumavských povodí. Ochrana přírody, 1, 35-38, available at: file:///C:/Users/ghisi.t/Downloads/OP_16-01_preview_12-1.pdf (last access: 11 September 2022).

Ide, J.I., Finér, L., Laurén, A., Piirainen, S., Launiainen, S., 2013. Effects of clear-cutting on annual and seasonal runoff from a boreal forest catchment in eastern Finland. For. Ecol. Manage. 304, 482–491. https://doi.org/10.1016/j.foreco.2013.05.051.

Iritz, Z., Lindroth, A., 1996. Energy partitioning in relation to leaf area development of short-rotation willow coppice. Agric. For. Meteorol. 81 (1–2), 119–130. https://doi.org/10.1016/0168-1923(95)02306-2.

Jiang, Z., Huete, A.R., Didan, K., Miura, T., 2008. Development of a two-band enhanced vegetation index without a blue band. Remote Sens. Environ. 112 (10), 3833–3845. https://doi.org/10.1016/j.rse.2008.06.006.

Kang, Y., Özdoğan, M., Zipper, S.C., Román, M.O., Walker, J., Hong, S.Y., Loheide, S.P., 2016. How universal is the relationship between remotely sensed vegetation indices and crop leaf area index? A global assessment. Remote sensing 8 (7), 597. https://doi.org/10.3390/rs8070597.

Kljun, N., Calanca, P., Rotach, M.W., Schmid, H.P., 2015. A simple two-dimensional parameterisation for Flux Footprint Prediction (FFP). Geosci. Model Dev. 8 (11), 3695–3713. https://doi.org/10.5194/gmd-8-3695-2015.

Kopáček, J., Bače, R., Hejzlar, J., Kaňa, J., Kučera, T., Matějka, K., Turek, J., 2020. Changes in microclimate and hydrology in an unmanaged mountain forest catchment after insect-induced tree dieback. Sci. Total Environ. 720, 137518 https://doi.org/10.1016/j.scitotenv.2020.137518.

Kowalska, N., Šigut, L., Stojanović, M., Fischer, M., Kyselova, I., Pavelka, M., 2020. Analysis of floodplain forest sensitivity to drought. Philos. Trans. R. Soc. B 375 (1810), 20190518. https://doi.org/10.1098/rstb.2019.0518.

Kubota, T., Kagawa, A., Abe, T., Hosoda, I., 2021. Effects of clear-cutting, meteorological, and physiological factors on evapotranspiration in the Kamabuchi experimental watershed in northern Japan. Hydrol. Process. 35 (4), e14111.

Kuraji, K., Gomyo, M., Nainar, A., 2019. Thinning of cypress forest increases subsurface runoff but reduces peak storm-runoff: A lysimeter observation. Hydrol. Res. Lett. 13, 49–54. https://doi.org/10.3178/hrl.13.49.

Larcher, W. (2003). Physiological plant ecology: ecophysiology and stress physiology of functional groups. Springer Science & Business Media, ISBN: 978-3-540-43516-7.

Liebert, R., Huntington, J., Morton, C., Sueki, S., Acharya, K., 2016. Reduced evapotranspiration from leaf beetle induced tamarisk defoliation in the Lower Virgin River using satellite-based energy balance. Ecohydrology 9 (1), 179–193. https://doi.org/10.1016/j.jnc.2013.08.00210.1002/eco.1623.

López, B. C., Gracia, C. A., Sabaté, S., & Keenan, T. (2009). Assessing the resilience of Mediterranean holm oaks to disturbances using selective thinning. *Acta Oecologica*, *35*(6), 849-854, 10.1016/j.jnc.2013.08.00210.1016/j.actao.2009.09.001.

Miklín, J., Čížek, L., 2014. Erasing a European biodiversity hot-spot: open woodlands, veteran trees and mature forests succumb to forestry intensification, succession, and logging in a UNESCO Biosphere Reserve. J. Nat. Conserv. 22 (1), 35–41. https://doi.org/10.1016/j.jnc.2013.08.002.

Mkhwanazi, M.M., & Chávez, J.L. (2012). Using METRIC to estimate surface energy fluxes over an alfalfa field in Eastern Colorado (Doctoral dissertation, Colorado State University. Libraries), available at: http://hydrologydays.colostate.edu/Papers_2012/Mcebisi_paper.pdf (last access: 11 September 2022).

Norman, J.M., Anderson, M.C., Kustas, W.P., French, A.N., Mecikalski, J.O.H.N., Torn, R., Tanner, B.C.W., 2003. Remote sensing of surface energy fluxes at 101-m pixel resolutions. Water Resour. Res. 39 (8) https://doi.org/10.1029/2002WR001775.

Oulehle, F., Fischer, M., Hruška, J., Chuman, T., Krám, P., Navrátil, T., Trnka, M., 2021. The GEOMON network of Czech catchments provides long-term insights into altered forest biogeochemistry: From acid atmospheric deposition to climate change. Hydrol. Process. 35 (5), e14204.

Pinheiro, J.C., Bates, D.M. (2000). Mixed-effects models in S and S-Plus.,(Springer-Verlag: New York). Mixed-Effects Models in S and S-Plus, ISBN: 978-0-387-22747-4.

R Core Team (2022). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL https://www.R-project.org/ (last access: 11 September 2022).

Roleček, J., Řepka, R., 2020. Formerly coppiced old growth stands act as refugia of threatened biodiversity in a managed steppic oak forest. For. Ecol. Manage. 472, 118245 https://doi.org/10.1016/j.foreco.2020.118245.

Řehoř, J., Brázdil, R., Trnka, M., Řezníčková, L., Balek, J., Možný, M., 2020. Regional effects of synoptic situations on soil drought in the Czech Republic. Theor. Appl. Climatol. 141 (3), 1383–1400. https://doi.org/10.1007/s00704-020-03275-4.

RStudio Team (2020). RStudio: Integrated Development for R. RStudio, PBC, Boston, MA URL http://www.rstudio.com/ (last access: 11 September 2022).

Senay, G.B., Budde, M., Verdin, J.P., Melesse, A.M., 2007. A coupled remote sensing and simplified surface energy balance approach to estimate actual evapotranspiration from irrigated fields. Sensors 7 (6), 979–1000. https://doi.org/10.3390/s7060979.

Stasik, R., Sojka, M., Korytowski, M. (2020). Assessment of the impact of clear-cutting on groundwater regime in swampy habitats. Baltic Forestry, 26(2), 10.46490/BF151.

Sun, G., Noormets, A., Gavazzi, M.J., McNulty, S.G., Chen, J., Domec, J.C., Skaggs, R.W., 2010. Energy and water balance of two contrasting loblolly pine plantations on the lower coastal plain of North Carolina, USA. For. Ecol. Manage. 259 (7), 1299–1310. https://doi.org/10.1016/j.foreco.2009.09.016.

Sun, X., Onda, Y., Otsuki, K., Kato, H., Gomi, T., Liu, X., 2017. Change in evapotranspiration partitioning after thinning in a Japanese cypress plantation. Trees-Structure and Function 31, 1411–1421. https://doi.org/10.1007/s00468-017-1555-1.

Thompson, C., Devito, K.J., Mendoza, C.A., 2018. Hydrologic impact of aspen harvesting within the subhumid Boreal Plains of Alberta. Hydrol. Process. 32 (26), 3924–3937. https://doi.org/10.1002/hyp.13301.

Trezza, R., Allen, R.G., Tasumi, M., 2013. Estimation of actual evapotranspiration along the Middle Rio Grande of New Mexico using MODIS and landsat imagery with the METRIC model. Remote Sens. (Basel) 5 (10), 5397–5423. https://doi.org/10.3390/rs5105397.

Trnka, M., Hlavinka, P., Možný, M., Semerádová, D., Štěpánek, P., Balek, J., Žalud, Z., 2020. Czech Drought Monitor System for monitoring and forecasting agricultural drought and drought impacts. Int. J. Climatol. 40 (14), 5941–5958. https://doi.org/10.1002/joc.6557.

U.S. Geological Survey, (2022a), Landsat missions, Landsat Collection 1, accessed June 30, 2022 at URL https://www.usgs.gov/landsat-missions/landsat-collection-1 (last access: 11 September 2022).

U.S. Geological Survey, (2022b) Earth Resources Observation and Science (EROS) Center, USGS EROS Archive - Landsat Archives - Landsat 8-9 OLI/TIRS Collection 2 Level-2 Science Products, accessed June 29, 2022 at URL https://www.usgs.gov/centers/eros/science/usgs-eros-archive-landsat-archives-landsat-8-9-olitirs-collection-2-level-2 (last access: 11 September 2022).

U.S. Geological Survey, (2022c), Earth Resources Observation and Science (EROS) Center, USGS EROS Archive - Digital Elevation - Shuttle Radar Topography Mission (SRTM) 1 Arc-Second Global, accessed April 16, 2022 at URL https://www.usgs.gov/centers/eros/science/usgs-eros-archive-digital-elevation-shuttle-radar-topography-mission-srtm-1?qt-science_center_objects=0#qt-science_center_objects (last access: 11 September 2022).

U.S. Geological Survey, (2022d), Earth Explorer, accessed April 16, 2022 at URL https://earthexplorer.usgs.gov/ (last access: 11 September 2022).

Venables, W.N., Ripley, B.D., 2002. Modern Applied Statistics with S, 4th ed. Springer. [Crossref], [Google Scholar], New York.

Westfall, P.H., 1997. Multiple testing of general contrasts using logical constraints and correlations. J. Am. Stat. Assoc. 92 (437), 299–306. https://doi.org/10.2307/2291474.

Yang, C.L., 2019. Effects of forest thinning on the long-term runoff changes of coniferous Forest plantation. Water 11, 2301. https://doi.org/10.3390/w11112301.

Yang, Y., Anderson, M.C., Gao, F., Hain, C.R., Semmens, K.A., Kustas, W.P., Sun, G., 2017. Daily Landsat-scale evapotranspiration estimation over a forested landscape in North Carolina, USA, using multi-satellite data fusion. Hydrol. Earth Syst. Sci. 21 (2), 1017–1037. https://doi.org/10.5194/hess-21-1017-2017.

Yang, Y., Anderson, M., Gao, F., Hain, C., Noormets, A., Sun, G., Sun, L., 2020. Investigating impacts of drought and disturbance on evapotranspiration over a forested landscape in North Carolina, USA using high spatiotemporal resolution remotely sensed data. Remote Sens. Environ. 238, 111018 https://doi.org/10.1016/j.rse.2018.12.017.

Zahradníček, P., Brázdil, R., Štěpánek, P., Trnka, M., 2021. Reflections of global warming in trends of temperature characteristics in the Czech Republic, 1961–2019. Int. J. Climatol. 41 (2), 1211–1229. https://doi.org/10.1002/joc.6791.

Zenone, T., Fischer, M., Arriga, N., Broeckx, L.S., Verlinden, M.S., Vanbeveren, S., Ceulemans, R., 2015. Biophysical drivers of the carbon dioxide, water vapor, and energy exchanges of a short-rotation poplar coppice. Agric. For. Meteorol. 209, 22–35. https://doi.org/10.1016/j.agrformet.2015.04.009.

Zhang, H., Anderson, R.G., Wang, D., 2015. Satellite-based crop coefficient and regional water use estimates for Hawaiian sugarcane. Field Crop Res. 180, 143-. https://doi.org/10.1016/j.fcr.2015.05.023.

Update

Forest Ecology and Management

Volume 556, Issue , 15 March 2024, Page

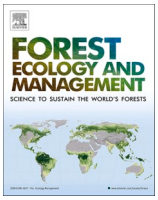
DOI: https://doi.org/10.1016/j.foreco.2024.121746

Forest Ecology and Management 556 (2024) 121746

Contents lists available at ScienceDirect

Forest Ecology and Management

journal homepage: www.elsevier.com/locate/foreco



Corrigendum

Corrigendum to "Faster evapotranspiration recovery compared to canopy development post clearcutting in a floodplain forest" [Forest Ecol. Manag. 532 (2023), 1–13/ 120828]

T. Ghisi [a,b,*], M. Fischer [a,b], N. Kowalska [a], G. Jocher [a], M. Orság [a,b], M. Bláhová [a,b], H. Nieto [c], L. Homolová [a], Z. Žalud [a,b], M. Trnka [a,b]

[a] Global Change Research Institute of the Czech Academy of Sciences, Bělidla 986/4b, 603 00 Brno, Czech Republic
[b] Mendel University in Brno, Institute of Agrosystems and Bioclimatology, Zemědělská 1, 613 00 Brno, Czech Republic
[c] Institute of Agricultural Sciences-CSIC, Serrano 115 b, 28006 Madrid, Spain

The authors regret that Fig. 8 and Fig. 9 are identical. The correct Fig. 8 is displayed below.

The corresponding author would like to apologize for any inconvenience caused.

DOI of original article: https://doi.org/10.1016/j.foreco.2023.120828.
* Corresponding author at: Global Change Research Institute of the Czech Academy of Sciences, Bělidla 986/4b, 603 00 Brno, Czech Republic.
E-mail address: ghisi.t@czechglobe.cz (T. Ghisi).

https://doi.org/10.1016/j.foreco.2024.121746

Available online 6 February 2024
0378-1127/© 2023 Elsevier B.V. All rights reserved.

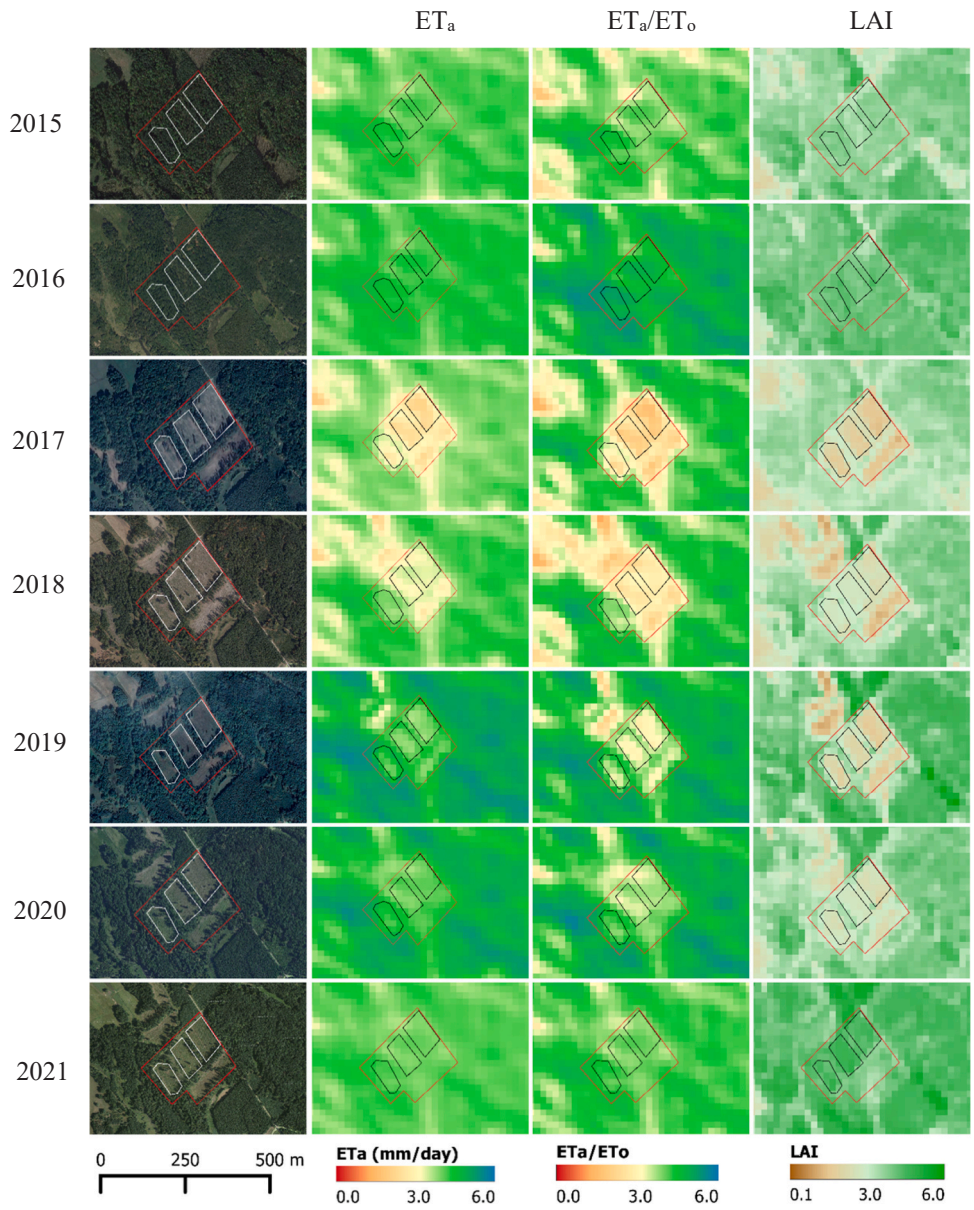


Fig. 8. Selected ET_a, ET_a/ET_o and LAI summer images in the 2015–2021 period of AOI 2. The red line is the border of AOI 2. The black or white line is the border of one clearcutting area. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).